

MacroTaylor: Rank-2 Laplace Approximation for the Posterior over p (Moffatt 2007 scaling convention)

This note is a rewrite of `MacroTaylor_2_chatgpt_reseach.tex` in the *Moffatt (2007) scaling convention* for the prior and posterior covariances. The mathematics is identical; the only change is the choice of which object carries the explicit $1/N_{\text{ch}}$ factor.

Scaling convention. Following Moffatt (2007), we represent the prior and posterior covariances over p by their *per-channel* (intensive) scaled counterparts Σ and Σ_{post} , which are independent of the channel count N_{ch} . The *actual* prior and posterior covariances carry an explicit $1/N_{\text{ch}}$:

$$p \sim \mathcal{N}\left(\mu, \frac{1}{N_{\text{ch}}} \Sigma\right), \quad p \mid y \approx \mathcal{N}\left(p^*, \frac{1}{N_{\text{ch}}} \Sigma_{\text{post}}\right).$$

Adding channels makes the actual covariance shrink as $1/N_{\text{ch}}$, while Σ stays fixed. This is the natural convention for an ensemble of N_{ch} i.i.d. channels and reduces cleanly to the single-channel formulas of Moffatt (2007) when $\sigma^2 = 0$.

The **main scheme** is the *rank-2* Laplace approximation based on the *exact* Gaussian likelihood (including the $\frac{1}{2} \log V$ term). As a *pedagogical limit* and for connection to EKF-style approximations, we also present a *rank-1* “quasi-likelihood” scheme in which the $\log V$ term is dropped.

The goals are:

- to define the exact Gaussian likelihood and Moffatt-scaled prior in the macroscopic coordinates p ;
- to derive the exact gradient and Hessian of the negative log-posterior, and show that the Hessian is a rank-2 update of the (per-channel) prior precision Σ^{-1} ;
- to obtain a closed-form expression for the *scaled Laplace covariance* Σ_{post} via a rank-2 Woodbury inversion;

- to derive a mean-update (MAP) formula in which the $1/N_{\text{ch}}$ factors cancel and no explicit Σ^{-1} survives;
- to show how the rank-1 (no $\log V$) scheme appears as a limiting case and how it connects to quasi-likelihood/EKF practice.

Coordinates, simplex and singular covariance. The macroscopic occupancy vector p lives on the simplex

$$\mathcal{S} = \left\{ p \in \mathbb{R}^{1 \times d} : p_i \geq 0, \sum_i p_i = 1 \right\},$$

a $(d-1)$ -dimensional manifold embedded in $\mathbb{R}^{1 \times d}$. The scaled prior covariance Σ is therefore singular as a $d \times d$ matrix in ambient coordinates. Formally, one may view Σ^{-1} as the inverse on the intrinsic $(d-1)$ -dimensional subspace, or as a pseudo-inverse. In what follows:

- we freely use Σ^{-1} *algebraically* when applying Sherman–Morrison–Woodbury;
- all *implementable* expressions for the mean and covariance are written purely in terms of Σ (no explicit inverse);
- the log-determinant of Σ (or its pseudo-determinant) is a constant in p and does not affect the MAP.

0. Setup and Notation

We consider:

- Macroscopic occupancy vector

$$p \in \mathbb{R}^{1 \times d}$$

(row vector of state probabilities).

- Per-state conductance (or current) vector

$$\gamma \in \mathbb{R}^{d \times 1}$$

(column vector).

- Per-state intrinsic-variance vector

$$\sigma^2 \in \mathbb{R}^{d \times 1}.$$

The prior over p (restricted to the simplex) is Gaussian in Moffatt-scaled form:

$$p \sim \mathcal{N}\left(\mu, \frac{1}{N_{\text{ch}}} \Sigma\right),$$

where $\mu \in \mathbb{R}^{1 \times d}$ is the prior mean and $\Sigma \in \mathbb{R}^{d \times d}$ is the *scaled* (per-channel) prior covariance. The corresponding prior precision is

$$\Lambda_p = \left(\frac{1}{N_{\text{ch}}} \Sigma\right)^{-1} = N_{\text{ch}} \Sigma^{-1}.$$

Over an interval we observe a scalar averaged current y which, for a given p , we model as Gaussian:

$$y \mid p \sim \mathcal{N}(\mu_y(p), V(p)),$$

with

$$\mu_y(p) = N_{\text{ch}}(p \cdot \gamma), \quad V(p) = \epsilon^2 + N_{\text{ch}}(p \cdot \sigma^2),$$

where ϵ^2 is the extrinsic (instrument) variance and $N_{\text{ch}}(p \cdot \sigma^2)$ is the intrinsic interval variance from channel stochasticity.

Define the residual

$$\delta(p) = y - \mu_y(p) = y - N_{\text{ch}}(p \cdot \gamma).$$

We will also use the *tilted* vector

$$v(p) = \gamma + \frac{\delta(p)}{V(p)} \sigma^2 \in \mathbb{R}^{d \times 1},$$

which will factor the Hessian.

1. Exact Gaussian Likelihood and Moffatt-scaled Prior

We start from the standard Gaussian forms and derive the exact negative log-posterior. We then introduce an “energy” that is twice the negative log-posterior, which simplifies later algebra.

1.1 Exact likelihood

Given p , the likelihood is

$$p(y \mid p) = \frac{1}{\sqrt{2\pi V(p)}} \exp\left(-\frac{1}{2} \frac{\delta(p)^2}{V(p)}\right).$$

The log-likelihood is

$$\log p(y | p) = -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log V(p) - \frac{1}{2} \frac{\delta(p)^2}{V(p)}.$$

Up to an additive constant $-\frac{1}{2} \log(2\pi)$, the *negative* log-likelihood is

$$F_{\text{like}}(p) = \frac{1}{2} \log V(p) + \frac{1}{2} \frac{\delta(p)^2}{V(p)} \quad (+ \text{ const}).$$

1.2 Moffatt-scaled prior

The Moffatt-scaled prior density is

$$p(p) \propto \exp\left(-\frac{1}{2}(p - \mu) (N_{\text{ch}} \Sigma^{-1}) (p - \mu)^\top\right),$$

since the actual prior covariance is $\frac{1}{N_{\text{ch}}} \Sigma$ and its precision is $N_{\text{ch}} \Sigma^{-1}$. The corresponding negative log-prior (up to a constant) is

$$F_{\text{prior}}(p) = \frac{N_{\text{ch}}}{2} (p - \mu) \Sigma^{-1} (p - \mu)^\top \quad (+ \text{ const}).$$

The log-determinant term $\frac{1}{2} \log \det\left(\frac{1}{N_{\text{ch}}} \Sigma\right)$ is a constant in p and therefore does not affect the MAP; we drop it throughout. However, we should note that the posterior covariance does depend on p , so this is actually a simplification to make the calculations manageable.

1.3 Exact negative log-posterior and energy

The (unnormalised) posterior is

$$p(p | y) \propto p(y | p) p(p),$$

so the exact negative log-posterior (up to a constant) is

$$F_{\text{exact}}(p) = F_{\text{like}}(p) + F_{\text{prior}}(p).$$

Explicitly:

$$F_{\text{exact}}(p) = \frac{1}{2} \log V(p) + \frac{1}{2} \frac{\delta(p)^2}{V(p)} + \frac{N_{\text{ch}}}{2} (p - \mu) \Sigma^{-1} (p - \mu)^\top \quad (+ \text{ const}).$$

We will minimise $F_{\text{exact}}(p)$ to obtain the exact MAP. For algebraic convenience, define the associated “energy”

$$E_{\text{exact}}(p) := 2F_{\text{exact}}(p).$$

Up to an additive constant,

$$E_{\text{exact}}(p) = \underbrace{\log V(p) + \frac{\delta(p)^2}{V(p)}}_{L(p)+M(p)} + \underbrace{N_{\text{ch}}(p-\mu)\Sigma^{-1}(p-\mu)^\top}_{P(p)}.$$

Thus:

$$E_{\text{exact}}(p) = L(p) + M(p) + P(p).$$

Multiplying by a positive constant does not change the MAP; moreover, if we write $\mathbf{H}(p) = \nabla^2 E_{\text{exact}}(p)$, then the *actual* Laplace covariance of the posterior is

$$\frac{1}{N_{\text{ch}}} \Sigma_{\text{Lap}} \approx [\nabla^2 F_{\text{exact}}(p^\star)]^{-1} = 2\mathbf{H}(p^\star)^{-1},$$

where $p^\star$ is the MAP and Σ_{Lap} denotes the *scaled* (per-channel) Laplace covariance.

2. Gradient and Hessian of the Exact Energy

We now derive the gradient and Hessian of $E_{\text{exact}}(p)$. We keep track of three contributions:

$$L(p) = \log V(p), \quad M(p) = \frac{\delta(p)^2}{V(p)}, \quad P(p) = N_{\text{ch}}(p-\mu)\Sigma^{-1}(p-\mu)^\top.$$

Here p is a row vector, γ and σ^2 are column vectors.

Recall:

$$\delta(p) = y - N_{\text{ch}}(p \cdot \gamma), \quad V(p) = \epsilon^2 + N_{\text{ch}}(p \cdot \sigma^2).$$

Then

$$\nabla \delta(p) = -N_{\text{ch}}\gamma^\top, \quad \nabla V(p) = N_{\text{ch}}(\sigma^2)^\top,$$

both row vectors, and $\nabla^2 \delta = 0$, $\nabla^2 V = 0$.

2.1 Prior gradient and Hessian

Write

$$P(p) = (p - \mu) \Lambda_p (p - \mu)^\top, \quad \Lambda_p = N_{\text{ch}} \Sigma^{-1}.$$

Since Λ_p is symmetric,

$$\nabla P(p) = 2(p - \mu) \Lambda_p = 2N_{\text{ch}}(p - \mu) \Sigma^{-1},$$

and

$$\nabla^2 P(p) = 2\Lambda_p = 2N_{\text{ch}} \Sigma^{-1}.$$

2.2 Likelihood gradient: $M(p)$ and $L(p)$

For

$$M(p) = \frac{\delta(p)^2}{V(p)},$$

the gradient (using the quotient rule) is

$$\nabla M(p) = \frac{2\delta}{V} \nabla \delta - \frac{\delta^2}{V^2} \nabla V,$$

so

$$\nabla M(p) = \frac{2\delta}{V} (-N_{\text{ch}} \gamma^\top) - \frac{\delta^2}{V^2} N_{\text{ch}} (\sigma^2)^\top = -N_{\text{ch}} \left[\frac{2\delta}{V} \gamma^\top + \frac{\delta^2}{V^2} (\sigma^2)^\top \right].$$

For

$$L(p) = \log V(p),$$

$$\nabla L(p) = \frac{1}{V} \nabla V = \frac{N_{\text{ch}}}{V} (\sigma^2)^\top.$$

Thus the *exact* gradient of $E_{\text{exact}} = L + M + P$ is

$$\begin{aligned} \mathbf{g}_{\text{exact}}(p) &= \nabla E_{\text{exact}}(p) = \nabla P(p) + \nabla M(p) + \nabla L(p) \\ &= 2N_{\text{ch}}(p - \mu) \Sigma^{-1} - N_{\text{ch}} \left[\frac{2\delta}{V} \gamma^\top + \frac{\delta^2}{V^2} (\sigma^2)^\top \right] + \frac{N_{\text{ch}}}{V} (\sigma^2)^\top. \end{aligned}$$

Grouping $(\sigma^2)^\top$ terms:

$$\mathbf{g}_{\text{exact}}(p) = 2N_{\text{ch}}(p - \mu) \Sigma^{-1} - N_{\text{ch}} \left[\frac{2\delta(p)}{V(p)} \gamma^\top + \left(\frac{\delta(p)^2}{V(p)^2} - \frac{1}{V(p)} \right) (\sigma^2)^\top \right].$$

2.3 Likelihood Hessian: $M(p)$ and $L(p)$

The Hessian of $M(p)$ was derived in detail in the rank-1 setting and does not change:

$$\nabla^2 M(p) = 2N_{\text{ch}}^2 \left[\frac{1}{V} \gamma \gamma^\top + \frac{\delta}{V^2} (\gamma(\sigma^2)^\top + \sigma^2 \gamma^\top) + \frac{\delta^2}{V^3} \sigma^2 \sigma^{2\top} \right].$$

For $L(p) = \log V(p)$,

$$\nabla^2 L(p) = -\frac{1}{V^2} (\nabla V)^\top \nabla V = -\frac{N_{\text{ch}}^2}{V^2} \sigma^2 \sigma^{2\top}.$$

So

$$\nabla^2 (M(p) + L(p)) = 2N_{\text{ch}}^2 \left[\frac{1}{V} \gamma \gamma^\top + \frac{\delta}{V^2} (\gamma(\sigma^2)^\top + \sigma^2 \gamma^\top) + \left(\frac{\delta^2}{V^3} - \frac{1}{2V^2} \right) \sigma^2 \sigma^{2\top} \right].$$

2.4 Full exact Hessian and rank-2 structure

Adding the prior Hessian:

$$\nabla^2 P(p) = 2N_{\text{ch}} \Sigma^{-1},$$

we obtain

$$\mathbf{H}_{\text{exact}}(p) := \nabla^2 E_{\text{exact}}(p) = 2N_{\text{ch}} \Sigma^{-1} + 2N_{\text{ch}}^2 \left[\frac{1}{V} \gamma \gamma^\top + \frac{\delta}{V^2} (\gamma(\sigma^2)^\top + \sigma^2 \gamma^\top) + \left(\frac{\delta^2}{V^3} - \frac{1}{2V^2} \right) \sigma^2 \sigma^{2\top} \right].$$

It is convenient to express this as a rank-2 update of the per-channel prior precision. Define

$$v(p) = \gamma + \frac{\delta(p)}{V(p)} \sigma^2.$$

Set

$$u_1(p) = v(p), \quad u_2(p) = \sigma^2.$$

We seek a representation

$$\mathbf{H}_{\text{exact}}(p) = 2N_{\text{ch}} [\Sigma^{-1} + U(p) C(p) U(p)^\top],$$

with

$$U(p) = [u_1(p), u_2(p)] \in \mathbb{R}^{d \times 2},$$

and $C(p)$ a diagonal 2×2 matrix.

From the rank-1 analysis without $\log V$ we know that

$$\frac{N_{\text{ch}}}{V} vv^\top = \frac{N_{\text{ch}}}{V} u_1 u_1^\top$$

reconstructs the $\gamma\gamma^\top$, $\gamma(\sigma^2)^\top$ and $\sigma^2\sigma^{2\top}$ terms in M alone. The extra contribution from L is

$$-\frac{N_{\text{ch}}^2}{V^2} \sigma^2 \sigma^{2\top} = 2N_{\text{ch}} \left(-\frac{N_{\text{ch}}}{2V^2} \right) u_2 u_2^\top.$$

Thus we can take

$$C(p) = \begin{pmatrix} \alpha(p) & 0 \\ 0 & \beta(p) \end{pmatrix}, \quad \alpha(p) = \frac{N_{\text{ch}}}{V(p)}, \quad \beta(p) = -\frac{N_{\text{ch}}}{2V(p)^2},$$

so that

$$U(p)C(p)U(p)^\top = \alpha(p)u_1u_1^\top + \beta(p)u_2u_2^\top.$$

We have therefore:

$$\boxed{\mathbf{H}_{\text{exact}}(p) = 2N_{\text{ch}}[\Sigma^{-1} + U(p)C(p)U(p)^\top]},$$

a rank-2 update of the (per-channel) prior precision.

3. Rank-2 Woodbury, Exact Laplace Covariance and Mean

We now fix an expansion point p_0 and compute the Laplace covariance and mean using the rank-2 structure. In practice p_0 will be taken as the current Newton iterate or the final MAP $p^\star$.

3.1 Rank-2 Woodbury inversion

The Woodbury identity for a rank- k update says:

$$(A + UCU^\top)^{-1} = A^{-1} - A^{-1}U(C^{-1} + U^\top A^{-1}U)^{-1}U^\top A^{-1},$$

with A invertible, U of shape $d \times k$, and C of shape $k \times k$.

Formally, on the intrinsic $(d-1)$ -dimensional subspace we put

$$A = \Sigma^{-1}, \quad A^{-1} = \Sigma, \quad U = U(p_0), \quad C = C(p_0).$$

Then

$$(\Sigma^{-1} + UCU^\top)^{-1} = \Sigma - \Sigma UKU^\top \Sigma,$$

where

$$K = (C^{-1} + U^\top \Sigma U)^{-1} \in \mathbb{R}^{2 \times 2}.$$

At p_0 we have

$$\mathbf{H}_{\text{exact}}(p_0) = 2N_{\text{ch}} [\Sigma^{-1} + U(p_0)C(p_0)U(p_0)^\top],$$

so

$$\mathbf{H}_{\text{exact}}(p_0)^{-1} = \frac{1}{2N_{\text{ch}}} \left[\Sigma - \Sigma U(p_0)K(p_0)U(p_0)^\top \Sigma \right].$$

The exact *actual* Laplace covariance at p_0 is $2\mathbf{H}_{\text{exact}}(p_0)^{-1} = \frac{1}{N_{\text{ch}}} \Sigma_{\text{post}}^{\text{exact}}(p_0)$, where the *scaled* Laplace covariance is

$$\boxed{\Sigma_{\text{post}}^{\text{exact}}(p_0) = \Sigma - \Sigma U(p_0)K(p_0)U(p_0)^\top \Sigma.}$$

The actual posterior covariance is $\frac{1}{N_{\text{ch}}} \Sigma_{\text{post}}^{\text{exact}}(p_0)$ and shrinks as $1/N_{\text{ch}}$ as expected. $\Sigma_{\text{post}}^{\text{exact}}$ itself is N_{ch} -independent in structure.

Explicitly,

$$U(p_0) = [v(p_0), \sigma^2], \quad C(p_0) = \begin{pmatrix} \alpha & 0 \\ 0 & \beta \end{pmatrix},$$

with

$$\alpha = \frac{N_{\text{ch}}}{V(p_0)}, \quad \beta = -\frac{N_{\text{ch}}}{2V(p_0)^2},$$

and

$$K(p_0) = \left(C(p_0)^{-1} + U(p_0)^\top \Sigma U(p_0) \right)^{-1} \in \mathbb{R}^{2 \times 2}.$$

The 2×2 matrix

$$M = C^{-1} + U^\top \Sigma U$$

can be written in terms of scalar contractions

$$s = v^\top \Sigma v, \quad b = v^\top \Sigma \sigma^2, \quad c = \sigma^{2\top} \Sigma \sigma^2,$$

as

$$M = \begin{pmatrix} \alpha^{-1} + s & b \\ b & \beta^{-1} + c \end{pmatrix} = \begin{pmatrix} \frac{V(p_0)}{N_{\text{ch}}} + s & b \\ b & -\frac{2V(p_0)^2}{N_{\text{ch}}} + c \end{pmatrix}.$$

Then $K(p_0) = M^{-1}$ is obtained with the usual 2×2 inverse, either analytically or numerically.

Remark (singular prior). Again, Σ^{-1} appears only formally within the Woodbury identity. The implementable scaled covariance $\Sigma_{\text{post}}^{\text{exact}}(p_0)$ contains only Σ , $U(p_0)$ and the small 2×2 inverse $K(p_0)$.

3.2 Exact Laplace mean and elimination of Σ^{-1}

The Laplace mean at p_0 is

$$p_{\text{post}}^{\text{exact}} = p_0 - \mathbf{g}_{\text{exact}}(p_0) \mathbf{H}_{\text{exact}}(p_0)^{-1} = p_0 - \frac{1}{2} \mathbf{g}_{\text{exact}}(p_0) \left(\frac{1}{N_{\text{ch}}} \Sigma_{\text{post}}^{\text{exact}}(p_0) \right),$$

with $\mathbf{g}_{\text{exact}}$ from Section 2.2.

Introduce

$$\Delta p = p_0 - \mu$$

and write

$$\mathbf{g}_{\text{exact}}(p_0) = 2N_{\text{ch}} \Delta p \Sigma^{-1} - N_{\text{ch}} q^\top,$$

where

$$q^\top = \frac{2\delta}{V} \gamma^\top + \left(\frac{\delta^2}{V^2} - \frac{1}{V} \right) (\sigma^2)^\top,$$

with $\delta = \delta(p_0)$ and $V = V(p_0)$.

Then

$$\begin{aligned} p_{\text{post}}^{\text{exact}} &= p_0 - \frac{1}{2} (2N_{\text{ch}} \Delta p \Sigma^{-1} - N_{\text{ch}} q^\top) \left(\frac{1}{N_{\text{ch}}} \Sigma_{\text{post}}^{\text{exact}}(p_0) \right) \\ &= p_0 - \Delta p \Sigma^{-1} \Sigma_{\text{post}}^{\text{exact}}(p_0) + \frac{1}{2} q^\top \Sigma_{\text{post}}^{\text{exact}}(p_0). \end{aligned}$$

Note that the N_{ch} in the gradient cancels with the $1/N_{\text{ch}}$ in the actual posterior covariance; this is the characteristic simplification of the Moffatt scaling.

Using

$$\Sigma_{\text{post}}^{\text{exact}}(p_0) = \Sigma - \Sigma U K U^\top \Sigma,$$

we obtain

$$\begin{aligned} \Sigma^{-1} \Sigma_{\text{post}}^{\text{exact}}(p_0) &= \Sigma^{-1} (\Sigma - \Sigma U K U^\top \Sigma) \\ &= I - U K U^\top \Sigma, \end{aligned}$$

so

$$\Delta p \Sigma^{-1} \Sigma_{\text{post}}^{\text{exact}}(p_0) = \Delta p - \Delta p U K U^\top \Sigma.$$

Therefore

$$\begin{aligned} p_{\text{post}}^{\text{exact}} &= p_0 - \left[\Delta p - \Delta p U K U^\top \Sigma \right] + \frac{1}{2} q^\top \Sigma_{\text{post}}^{\text{exact}}(p_0) \\ &= (p_0 - \Delta p) + \Delta p U K U^\top \Sigma + \frac{1}{2} q^\top \Sigma_{\text{post}}^{\text{exact}}(p_0). \end{aligned}$$

Since $p_0 - \Delta p = \mu$, we arrive at the exact mean update with no explicit Σ^{-1} and no surviving N_{ch} factor:

$$\boxed{p_{\text{post}}^{\text{exact}} = \mu + \Delta p U(p_0) K(p_0) U(p_0)^\top \Sigma + \frac{1}{2} q^\top \Sigma_{\text{post}}^{\text{exact}}(p_0),}$$

with

$$\Delta p = p_0 - \mu, \quad q^\top = \frac{2\delta}{V} \gamma^\top + \left(\frac{\delta^2}{V^2} - \frac{1}{V} \right) (\sigma^2)^\top,$$

and $\Sigma_{\text{post}}^{\text{exact}}(p_0)$ given above.

The Moffatt scaling makes this update intuitive: the mean depends only on the per-channel scaled covariance Σ , and is independent of N_{ch} except through the residual $\delta(p_0)$ and the variance $V(p_0)$, which carry the actual data scale.

3.3 Newton–MAP scheme and MAP–Laplace posterior

The exact MAP is defined as

$$p^\star = \arg \min_{p \in \mathcal{S}} F_{\text{exact}}(p) = \arg \min_{p \in \mathcal{S}} E_{\text{exact}}(p),$$

subject to the simplex constraints. We locate $p^\star$ via Newton’s method applied to E_{exact} .

Let $p^{(k)}$ be the current iterate. One Newton step is

$$p^{(k+1)} = p^{(k)} - \Delta p^{(k)}, \quad \mathbf{H}_{\text{exact}}(p^{(k)}) \Delta p^{(k)\top} = \mathbf{g}_{\text{exact}}(p^{(k)})^\top.$$

Algorithmically:

1. Initialise with a feasible point, typically $p^{(0)} = \mu$.
2. For $k = 0, 1, 2, \dots$ until convergence:
 - (a) Evaluate $\delta^{(k)} = \delta(p^{(k)})$, $V^{(k)} = V(p^{(k)})$, $v^{(k)} = v(p^{(k)})$.
 - (b) Form $\mathbf{g}_{\text{exact}}(p^{(k)})$ and $\mathbf{H}_{\text{exact}}(p^{(k)})$ using the formulas above.

- (c) Compute the exact scaled Laplace covariance $\Sigma_{\text{post}}^{\text{exact}}(p^{(k)})$ using the rank-2 Woodbury formula.
- (d) Update the mean using the exact mean formula with $p_0 = p^{(k)}$:

$$p^{(k+1)} = \mu + \Delta p^{(k)} U^{(k)} K^{(k)} U^{(k)\top} \Sigma + \frac{1}{2} q^{(k)\top} \Sigma_{\text{post}}^{\text{exact}}(p^{(k)}),$$

where $\Delta p^{(k)} = p^{(k)} - \mu$, $U^{(k)} = U(p^{(k)})$ and $q^{(k)}$ is defined by $\delta^{(k)}$, $V^{(k)}$.

- (e) Optionally project $p^{(k+1)}$ back to the simplex (enforce $p_i^{(k+1)} \geq 0$ and renormalise).

After convergence, $p^{(k)} \rightarrow p^*$ and we define the MAP–Laplace posterior as

$$p \mid y \approx \mathcal{N}(p^*, \frac{1}{N_{\text{ch}}} \Sigma_{\text{Lap}}),$$

with the *scaled* Laplace covariance

$$\Sigma_{\text{Lap}} = \Sigma_{\text{post}}^{\text{exact}}(p^*) = \Sigma - \Sigma U(p^*) K(p^*) U(p^*)^\top \Sigma.$$

The actual posterior covariance is $\frac{1}{N_{\text{ch}}} \Sigma_{\text{Lap}}$ and inherits the $1/N_{\text{ch}}$ shrinkage of the Moffatt convention.

4. Rank-1 Quasi-Laplace Scheme as a Special Case

We now present the *quasi* (rank-1) scheme obtained by *dropping* the $\log V(p)$ term from the likelihood. This corresponds to a quasi-likelihood / weighted least-squares objective. It is useful pedagogically and connects to EKF-like approximations.

4.1 Quasi energy and Hessian

Define the quasi energy

$$E_{\text{quasi}}(p) = M(p) + P(p) = \frac{\delta(p)^2}{V(p)} + N_{\text{ch}} (p - \mu) \Sigma^{-1} (p - \mu)^\top.$$

The gradient is

$$\mathbf{g}_{\text{quasi}}(p) = 2N_{\text{ch}}(p - \mu)\Sigma^{-1} - N_{\text{ch}} \left[\frac{2\delta(p)}{V(p)} \gamma^\top + \frac{\delta(p)^2}{V(p)^2} (\sigma^2)^\top \right].$$

The Hessian is

$$\mathbf{H}_{\text{quasi}}(p) = 2N_{\text{ch}}\Sigma^{-1} + 2N_{\text{ch}}^2 \left[\frac{1}{V}\gamma\gamma^\top + \frac{\delta}{V^2} (\gamma(\sigma^2)^\top + \sigma^2\gamma^\top) + \frac{\delta^2}{V^3} \sigma^2\sigma^{2\top} \right].$$

Introduce

$$v(p) = \gamma + \frac{\delta(p)}{V(p)}\sigma^2, \quad u(p) = \sqrt{\frac{N_{\text{ch}}}{V(p)}} v(p).$$

Then

$$\mathbf{H}_{\text{quasi}}(p) = 2N_{\text{ch}} \left[\Sigma^{-1} + u(p)u(p)^\top \right].$$

This is a rank-1 update of the per-channel prior precision.

4.2 Rank-1 scaled covariance and mean

Applying Sherman–Morrison with $A = \Sigma^{-1}$ and $u = u(p_0)$ at an expansion point p_0 :

$$(\Sigma^{-1} + uu^\top)^{-1} = \Sigma - \frac{\Sigma uu^\top \Sigma}{1 + u^\top \Sigma u}.$$

Therefore

$$\mathbf{H}_{\text{quasi}}(p_0)^{-1} = \frac{1}{2N_{\text{ch}}} \left[\Sigma - \frac{\Sigma uu^\top \Sigma}{1 + u^\top \Sigma u} \right].$$

The actual quasi Laplace covariance at p_0 is $2\mathbf{H}_{\text{quasi}}(p_0)^{-1} = \frac{1}{N_{\text{ch}}} \Sigma_{\text{post}}^{\text{quasi}}(p_0)$, and the *scaled* quasi Laplace covariance is

$$\Sigma_{\text{post}}^{\text{quasi}}(p_0) = \Sigma - \frac{\Sigma uu^\top \Sigma}{1 + u^\top \Sigma u}.$$

Substituting $u = \sqrt{N_{\text{ch}}/V} v$ and defining $s = v^\top \Sigma v$, we obtain

$$\Sigma_{\text{post}}^{\text{quasi}}(p_0) = \Sigma - \frac{N_{\text{ch}}}{V(p_0) + N_{\text{ch}}s(p_0)} \Sigma v(p_0)v(p_0)^\top \Sigma.$$

This matches Eq. (15) of the gemmini-Moffatt note exactly.

The quasi mean at p_0 is

$$p_{\text{post}}^{\text{quasi}} = p_0 - \frac{1}{2} \mathbf{g}_{\text{quasi}}(p_0) \left(\frac{1}{N_{\text{ch}}} \Sigma_{\text{post}}^{\text{quasi}}(p_0) \right).$$

Writing $\Delta p = p_0 - \mu$ and using the identity

$$\Sigma^{-1} \Sigma_{\text{post}}^{\text{quasi}} = I - \frac{N_{\text{ch}}}{V + N_{\text{ch}}s} vv^\top \Sigma,$$

we obtain, after the same algebra as in Section 3.2 (with the N_{ch} cancelling against $1/N_{\text{ch}}$):

$$\boxed{p_{\text{post}}^{\text{quasi}} = \mu + \frac{N_{\text{ch}}}{V(p_0) + N_{\text{ch}}s(p_0)} (p_0 - \mu) v(p_0) v(p_0)^\top \Sigma + \frac{\delta(p_0)}{V(p_0)} \gamma^\top \Sigma_{\text{post}}^{\text{quasi}}(p_0) + \frac{\delta(p_0)^2}{2V(p_0)^2} (\sigma^2)^\top \Sigma_{\text{post}}^{\text{quasi}}(p_0)}.$$

Again, no explicit Σ^{-1} survives and no N_{ch} factor multiplies the scaled posterior covariance.

4.3 Special case $p_0 = \mu$ and connection to EKF

If we choose $p_0 = \mu$, then $\Delta p = 0$ and the first line of the quasi mean vanishes. In that case:

$$\mathbf{g}_{\text{quasi}}(\mu) = -N_{\text{ch}} \left[\frac{2\delta}{V} \gamma^\top + \frac{\delta^2}{V^2} (\sigma^2)^\top \right] = -\frac{N_{\text{ch}}\delta}{V} (\gamma^\top + v^\top),$$

and

$$p_{\text{post}}^{\text{quasi}} = \mu + \frac{\delta}{2V} (\gamma^\top + v^\top) \Sigma_{\text{post}}^{\text{quasi}},$$

where $v = \gamma + \frac{\delta}{V} \sigma^2$ and $\Sigma_{\text{post}}^{\text{quasi}}$ is evaluated at $p_0 = \mu$.

This is the form quoted in the deepseek-Moffatt note:

$$r_{\text{post}} = \mu + \frac{\delta}{2V} \Sigma_{\text{post}} (\gamma + v).$$

It can be interpreted as a single quasi-Laplace (or EKF-like) update from the prior mean, using only the mean-matching part of the likelihood and ignoring the $\log V$ term.

4.4 Recovery of Moffatt (2007): $\sigma^2 = 0$

When state-dependent intrinsic noise vanishes, $\sigma^2 = 0$, the tilted vector simplifies:

$$v = \gamma, \quad V = \epsilon^2,$$

and the quasi-Laplace formulas reduce to

$$\Sigma_{\text{post}}^{\text{quasi}}(p_0) = \Sigma - \frac{N_{\text{ch}}}{\epsilon^2 + N_{\text{ch}} \gamma^\top \Sigma \gamma} \Sigma \gamma \gamma^\top \Sigma,$$

$$p_{\text{post}}^{\text{quasi}}|_{p_0=\mu} = \mu + \frac{\delta}{\epsilon^2} \gamma^\top \Sigma_{\text{post}}^{\text{quasi}}.$$

These are exactly the single-step measurement-update equations of Moffatt (2007) (Eqs. 49–50 in the gemmini-Moffatt note’s numbering). The Moffatt scaling makes this recovery transparent: Σ , Σ_{post} and the mean update all reduce term-by-term to the 2007 expressions, with the $1/N_{\text{ch}}$ shrinkage living exclusively in the actual covariance of the Gaussian.

5. Summary and Interpretation

We now summarise the two schemes under Moffatt scaling.

5.1 Rank-2 (exact) scheme

- Likelihood:

$$p(y | p) = \mathcal{N}(N_{\text{ch}}(p \cdot \gamma), V(p)), \quad V(p) = \epsilon^2 + N_{\text{ch}}(p \cdot \sigma^2).$$

- Prior (Moffatt-scaled):

$$p \sim \mathcal{N}(\mu, \frac{1}{N_{\text{ch}}} \Sigma), \quad \Lambda_p = N_{\text{ch}} \Sigma^{-1}.$$

- Negative log-posterior (up to const):

$$F_{\text{exact}}(p) = \frac{1}{2} \log V(p) + \frac{1}{2} \frac{\delta(p)^2}{V(p)} + \frac{N_{\text{ch}}}{2} (p - \mu)^\top \Sigma^{-1} (p - \mu).$$

- Energy:

$$E_{\text{exact}}(p) = 2F_{\text{exact}}(p).$$

- Hessian:

$$\mathbf{H}_{\text{exact}}(p) = 2N_{\text{ch}} [\Sigma^{-1} + U(p)C(p)U(p)^\top],$$

with $U(p) = [v(p), \sigma^2]$ and diagonal $C(p) = \text{diag}(N_{\text{ch}}/V, -N_{\text{ch}}/(2V^2))$.

- Scaled covariance:

$$\Sigma_{\text{post}}^{\text{exact}}(p_0) = \Sigma - \Sigma U(p_0) K(p_0) U(p_0)^\top \Sigma,$$

with $K(p_0) = (C(p_0)^{-1} + U(p_0)^\top \Sigma U(p_0))^{-1}$. The actual posterior covariance is $\frac{1}{N_{\text{ch}}} \Sigma_{\text{post}}^{\text{exact}}(p_0)$.

- Mean:

$$p_{\text{post}}^{\text{exact}} = \mu + \Delta p U(p_0) K(p_0) U(p_0)^\top \Sigma + \frac{1}{2} q^\top \Sigma_{\text{post}}^{\text{exact}}(p_0).$$

- Newton iteration uses E_{exact} , converges to the true MAP p^* , and the scaled Laplace covariance at p^* is $\Sigma_{\text{Lap}} = \Sigma_{\text{post}}^{\text{exact}}(p^*)$.

This scheme fully incorporates the heteroscedastic variance $V(p)$ via both the δ^2/V term and the $\log V$ term of the Gaussian likelihood. Even when $\delta = 0$, the $\log V$ term can shift the MAP towards states with smaller predicted variance, in competition with the prior.

5.2 Rank-1 (quasi) scheme

- Energy:

$$E_{\text{quasi}}(p) = \frac{\delta(p)^2}{V(p)} + N_{\text{ch}} (p - \mu)^\top \Sigma^{-1} (p - \mu).$$

- Hessian:

$$\mathbf{H}_{\text{quasi}}(p) = 2N_{\text{ch}} \left[\Sigma^{-1} + \frac{N_{\text{ch}}}{V(p)} v(p)v(p)^\top \right],$$

a rank-1 update.

- Scaled covariance:

$$\Sigma_{\text{post}}^{\text{quasi}}(p_0) = \Sigma - \frac{N_{\text{ch}}}{V(p_0) + N_{\text{ch}} s(p_0)} \Sigma v(p_0)v(p_0)^\top \Sigma.$$

- Mean: the quasi mean $p_{\text{post}}^{\text{quasi}}$ is given by the rank-1 formula above, and in the special case $p_0 = \mu$ reduces to a single-step update

$$p_{\text{post}}^{\text{quasi}} = \mu + \frac{\delta}{2V} (\gamma^\top + v^\top) \Sigma_{\text{post}}^{\text{quasi}}.$$

This scheme ignores the $\log V(p)$ contribution to the likelihood and treats $V(p)$ only as a weight in the squared residual. It is analogous to using a weighted least-squares / quasi-likelihood objective in GLMs, and connects to EKF practice where a state-dependent variance is plugged into an effective R but not differentiated. When $\sigma^2 = 0$ it reduces to Moffatt (2007).

5.3 Practical recommendation

In MacroTaylor we take the rank-2 scheme as the *main method*:

- it is the correct Gaussian MAP for the interval model;
- the rank-2 Woodbury cost is negligible relative to the cost of computing the MacroIR quantities;
- the heteroscedastic variance $V(p)$ is physically meaningful and carries information about p .

The rank-1 scheme is retained as a *pedagogical device* and as a limiting case:

- it illustrates how a simpler quasi-likelihood/EKF update arises if one drops $\log V(p)$;
- it provides a useful check in regimes where $V(p)$ changes slowly with p or N_{ch} is large;
- with $\sigma^2 = 0$ it recovers Moffatt (2007) directly.

5.4 What the Moffatt scaling buys

Comparing this note with `MacroTaylor_2_chatgpt_research.tex` (which used the unscaled convention $p \sim \mathcal{N}(\mu, \Sigma_p)$ with $\Lambda_p = N_{\text{ch}} \Sigma_p^{-1}$):

- The scaled prior covariance Σ here is N_{ch} -independent; in the unscaled version, Σ_p played the same role but was conceptually the full population covariance. Numerically they coincide.
- The scaled posterior covariance Σ_{post} now has *no* explicit $1/N_{\text{ch}}$ factor in front; the $1/N_{\text{ch}}$ shrinkage lives in the Gaussian $\mathcal{N}(p^*, \frac{1}{N_{\text{ch}}} \Sigma_{\text{post}})$ rather than in Σ_{post} itself.
- The mean update has no $N_{\text{ch}}/2$ factor on the $q^\top \Sigma_{\text{post}}$ term: the N_{ch} in the gradient cancels with the $1/N_{\text{ch}}$ from the actual posterior covariance. Compare

$$p_{\text{post}}^{\text{exact}} = \mu + \Delta p U K U^\top \Sigma + \frac{1}{2} q^\top \Sigma_{\text{post}}^{\text{exact}}$$

(Moffatt scaling, this note) with

$$p_{\text{post}}^{\text{exact}} = \mu + \Delta p U K U^\top \Sigma_p + \frac{N_{\text{ch}}}{2} q^\top \Sigma_{\text{post}}^{\text{exact}}$$

(unscaled, original `MacroTaylor_2`); the two are algebraically equivalent under $\Sigma \leftrightarrow \Sigma_p$ and $\Sigma_{\text{post}} \leftrightarrow N_{\text{ch}} \Sigma_{\text{post}}^{\text{tex}}$.

- The limit $\sigma^2 \rightarrow 0$ reduces directly to the single-step update of Moffatt (2007), without any rescaling of either Σ or Σ_{post} .

The Moffatt scaling is therefore the natural choice when (i) comparing to the 2007 single-channel formulas, (ii) reasoning about the $N_{\text{ch}} \rightarrow \infty$ thermodynamic limit (in which only $\frac{1}{N_{\text{ch}}} \Sigma_{\text{post}}$ shrinks), and (iii) keeping all matrices in the algebra N_{ch} -independent in structure.